

Popularity Prediction of Music by Machine Learning Models

Zehao Xing*

School of International School, Beijing University of Posts and Telecommunications, Beijing, China

*Corresponding author: sonaldovski@bupt.edu.com

Abstract. Each piece of music has a considerable amount of information and attributes, and each listener has their own unique taste in music, with some songs being very popular and others being relatively niche. It becomes worthwhile to examine what kinds of music are more popular. It's clear that in this subject, the analysis and study of the available data is a "crucial first step". In this paper, we use several fitting models in machine learning as the theoretical basis, using pandas, sklearn, xgboost, and other related tools in python, to predict the popularity of music based on a dataset of music information originating from Kaggle. The most suitable machine learning model is founded for predicting music popularity and the effectiveness of its fit are evaluated. This study provides a methodological basis for finding the factors influencing music popularity in post-order studies and can be a key study in determining the factors that influence the popularity of music in the future.

Keywords: Music, xgboost, XGBRegressor, data.

1. Introduction

1.1. Background

In the music market, musicians are the ones who most want to know what their audience likes. Once know what the audience likes, it is easier to produce a more popular work. For tracks, there are a lot of features: some are acoustic in nature, such as beats, keys, etc., which are the nature of the track or the music itself. Some are non-acoustic and have to do with the emotion, attitude, etc. of the music, or even the composer or the album to which the music belongs.

Each listener, they have their own preferences, and for the public, some pieces are widely available, while others are less known. Discovering the public's preferences for the music itself will have a significant impact on the development of the music. And trying to find the relationship between the attributes of the music itself and the popularity of the music and trying to use these attributes to predict the popularity value of the music, is a very important part of this research.

1.2. Objection

The aim of this research is to find the most suitable model for predicting the popularity of music, as well as finding the relationship between the relevant properties of music and popularity, using the dataset used as a sample, to provide a methodological basis for future research. By finding the right models and methods, the further study can also explore the connection between certain properties and popularity and explore it in depth in future research.

In this paper, a music dataset which was obtained from the Kaggle [1] website is as a basis, and python as a tool to predict the popularity of music using different machine learning models, calculate and compare the relevant coefficients to evaluate the fit of the different models, and thus find the model that fits best. This study have used a python related data analysis tool to cleaned and trained the dataset, and found the most suitable model for the dataset so far, which is able to fit the effect of individual attributes on music popularity better [2].

2. Methodology

2.1. Source of data

The dataset for this study is from Kaggle [1] and records a dataset of 114,000 tuples and 19 attributes up to mid-2022 on music and its various attribute metrics, with popularity as the only dependent variable.

2.1.1 Dependent variables

In this study, the dependent variable is the outcome, which is the popularity of the data. Popularity is a value from 0-100, with 100 indicating the most popular. Popularity is calculated algorithmically, based on the total number of plays and the recent number of plays of the song. In general, songs with high total airplay are more popular, while songs with high recent airplay are more popular than songs with high past airplay.

2.1.2 Independent variables

There are many variables that have an impact on the popularity of music, such as instrumentals, loudness, etc. Some unwanted attributes have been removed, so only those that have an effect on the results are presented in Table 1:

Table 1. Variables of tracks

artists:	The artists' names who performed the track.
album_name	The album name in which the track appears
explicit	Whether or not the track has explicit lyrics. If yes, the value will be true, or it'll be false.
danceability	Danceability describes how suitable a track is for dancing with a value between 0 and 1. 1 means the most danceable
loudness	The overall loudness of a track in decibels (dB)
speechiness	Speechiness detects the presence of spoken words in a track with values in [0,1].
acousticness	A confidence measure from 0.0 to 1.0 of whether the track is acoustic.
instrumentalness	Predicts whether a track contains no vocals in value[0,1], 1.0 means that there is no vocal content in music
valence	A measure from 0.0 to 1.0 describes the musical positiveness conveyed by a track. Tracks with high valence sound more positive
time_signature	An estimated time signature. For example, 3 indicating time signature 3/4
track_genre	The genre in which the track belongs

Some of these variables are categorical variables, which need to transfer them into numeric variables. This be shown in the next part

2.2. Data processing

This study will use python to clean and analyze the data, the process of cleaning the data will be based on the pandas package under python. This data processing is divided into three main parts: Eliminating the duplicated and null tuples, Thermal coding and dataset split. The details are described as follows.

2.2.1 Eliminate the duplicated and null tuples

After checking the data by calling the function pandas, 577 duplicate items and no empty items were found, so the duplicate data were eliminated.

2.2.2 Thermal coding

By checking the attributes. It is found that there are several variable's data types are 'object', which is not proper for analyzing, so the solution is using the method of thermal coding to change them into numeric variables, as shown in Fig. 1.

```
def labelEncoder(df):  
    for c in df.columns:  
        if df[c].dtype=='object':  
            df[c]=df[c].fillna('N')  
            lbl=LabelEncoder()  
            lbl.fit(list(df[c].values))  
            df[c]=lbl.transform(df[c].values)  
    return df
```

Fig. 1 Thermal coding (Photo credit: Original)

2.2.3 dataset split

In this study, the argument 'test_size' was set as 0.2 to take one-sixth of all data as the training set.

2.3. machine learning models

This analysis uses a variety of algorithmic models for machine learning and compares their fitting results.

2.3.1 XGBoost

XGBoost is an open-source library for building machine learning models, particularly gradient-boosting models. It is designed to be highly efficient and flexible and is widely used in a variety of machine-learning applications. XGBoost is known for its ability to achieve high performance in many different types of machine-learning tasks [3].

Gradient boosting involves sequentially training several decision trees, with each tree attempting to fix the errors caused by the one before it. The projections of all the trees are combined to get the final prediction.

The gradient boosting algorithm is given several new characteristics by the XGBoost algorithm to increase its effectiveness and efficiency. These qualities consist of:

- Regularization: To avoid overfitting, XGBoost adds regularization to the model.
- Sparse data handling: Many machine learning tasks include sparse data, which XGBoost can handle effectively.
- Parallelization: XGBoost is substantially quicker than many other machine learning techniques since it can be parallelized to operate on several cores.
- Weighted quantile sketch: XGBoost makes predictions that are more precise by using a weighted quantile sketch to estimate the gradient and Hessian matrix distributions.

Overall, the xgboost algorithm can be expressed as:

$$\hat{y} = \sum_{k=1}^K f_k(x) \quad (1)$$

Where $\hat{y}$ is the final prediction, K is the number of decision trees, and $f_k(x)$ is the prediction of the k th decision tree [4].

2.3.2 Polynomial regression

Polynomial regression is a type of regression analysis. In polynomial regression, the relationship between the independent variable x and the dependent variable y is modeled as an n th-degree polynomial. It can be used to model relationships between variables that are not linear.

A polynomial regression model takes the form:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n \quad (2)$$

where β_0 is the intercept term, β_1 to β_n are the polynomial coefficients, and x is the independent variable [5].

2.3.3 Other learning models

Depending on the properties of the dataset itself, focusing on fitting the xgboost model and polynomial regression model is necessary. For reasons of accuracy and the need to find certain hidden properties, several methods such as linear regression [6], Ridge, Bayes Ridge[7], Lasso, etc.is added, in the hope of finding the most suitable fitted model

2.4. Evaluation Metrics

2.4.1 mean squared error

The mean squared error (MSE) is a commonly used loss function in machine learning that measures the average squared difference between the predicted values and the true values. It is defined as:

$$EMSE = \frac{1}{N \sum_{i=1}^N (d_i - y_i)^2} = \frac{1}{N \sum_{i=1}^N e_i^2} \quad (3)$$

Where N is training samples, di is the dimensions of vectors, yi is the output of the neural networks, ei is the sampling error [8].

2.4.2 R_Squared

R-squared, also known as the coefficient of determination, is a measure of how well a model fits the data. It is defined as the proportion of the variance in the dependent variable (y) that is explained by the independent variable (x). R_Squared is set in the interval [0,1], and can be calculated as shown in Fig. 2:

$$R_{m,M}^2 = 1 - \frac{\sum_{x=x_m}^{x_{m+1}-1} (y(x) - \hat{y}(x))^2}{\sum_{x=x_m}^{x_{m+1}-1} (y(x) - \bar{y})^2} \quad (4)$$

Where $\bar{y}$ is the average of y(x) [6].

2.4.3 Times

A time variable which in second is introduced to evaluate the running time of the program to find a more efficient model.

3. Results and Decision

3.1. Data visualization

3.1.1 visualization of dependent variables

By plotting the distribution of the populations, it can be found: population close to 0 accounts for more than 10% of all tracks, while the rest of the interval tends to be normally distributed, with a relatively large number of tracks in the 20-60 range. The distribution of popularity is shown in Fig. 2:

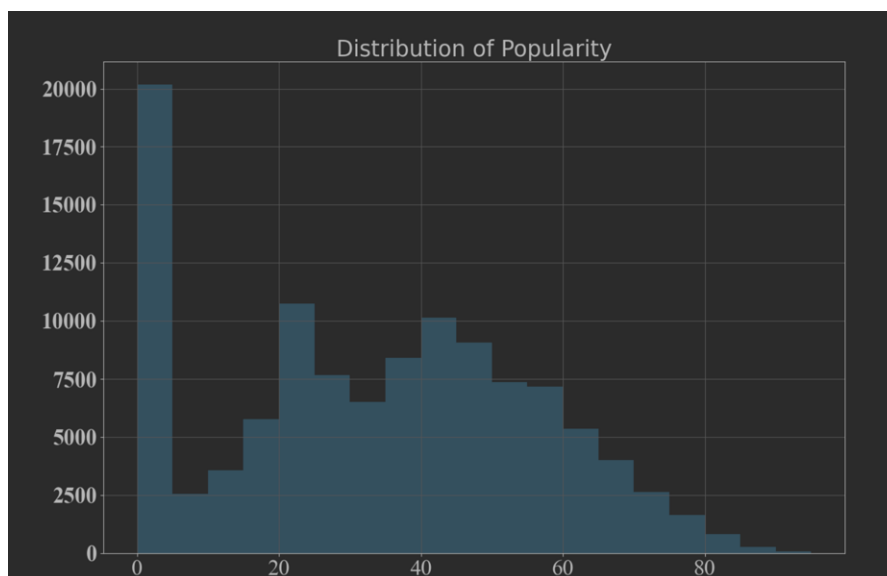


Fig. 2 Distribution of popularity (Photo credit: Original)

3.1.2 visualization of independent variables

This section used the pandas built-in describe() method and info() method to draw the following Table 2 to show the information of the independent variables.

Table 2. Types of all attributes

0	artists	object
1	album_name	object
2	track_name	object
3	popularity	int64
4	duration_ms	int64
5	explicit	bool
6	danceability	float64
7	energy	float64
8	key	int64
9	loudness	float64
10	mode	int64
11	speechiness	float64
12	acousticness	float64
13	instrumentalness	float64
14	liveness	float64
15	valence	float64
16	tempo	float64
17	time_signature	int64
18	track_genre	object

The values of three of these variables are converted from the object class (shown in Table 1): artists, album_name, and track_genre. The values of these three variables are listed in this table to show the results of the conversion, and the values are not statistically significant.

3.2. Variable Selection / Correlation analysis

In order to find the relationship between the independent and dependent variables and the relationship between the independent variables in order to select the appropriate variables, this section plotted heat maps for all variables for analysis, which is shown in Fig. 3:

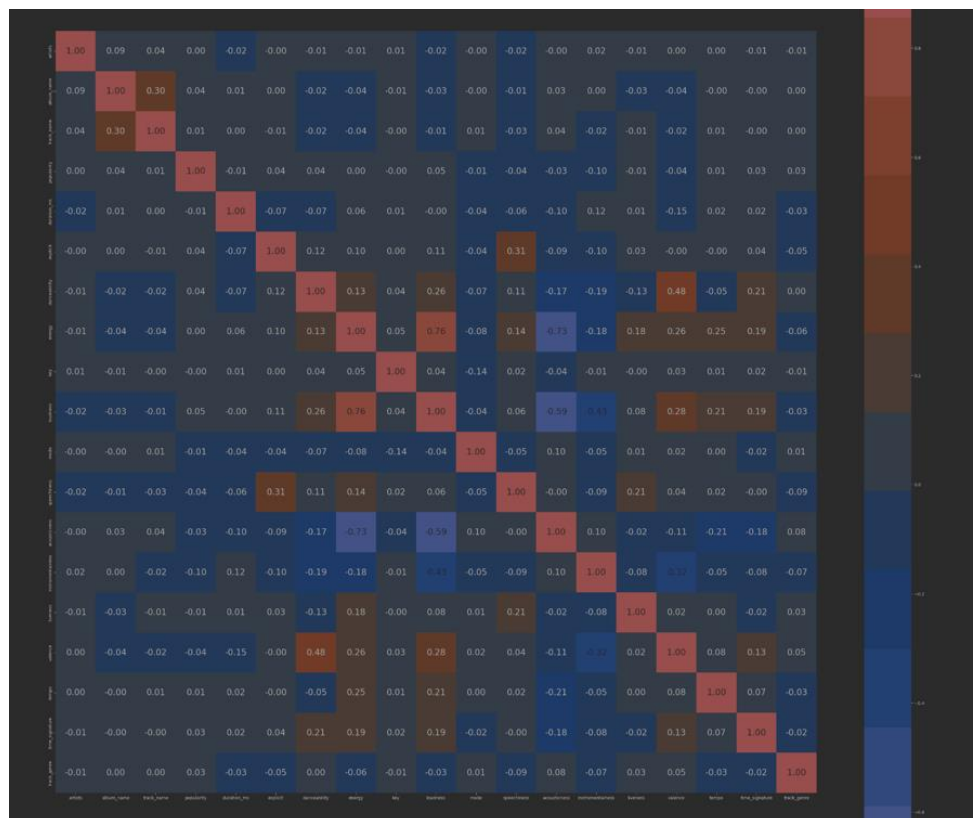


Fig. 3 Heatmap (The fourth column and fourth row represent popularity)

(Photo credit: Original)

In this heat map, the data in the fourth row and fourth column are data related to popularity. This section identified the variables with a very low correlation with popularity (≤ 0.01) and removed them. Then from the groups of variables with a high correlation (≥ 0.3), this section removed the one variable in each group that had a low correlation with popularity. During this process, 7 variables listed in Fig. 4 are removed and 11 variables remained, then the complexity of the analysis was simplified.

```
del music_data['track_name']
del music_data['duration_ms']
del music_data['energy']
del music_data['key']
del music_data['mode']
del music_data['liveness']
del music_data['tempo']
music_data
```

Fig. 4 Attributes that were deleted (Photo credit: Original)

3.3. Model training and evaluation

3.3.1 Model training and comparison

After the data had been selected and analyzed, this section trained the dataset separately using all the training models selected, calculated evaluation metrics for each model and visualized them.

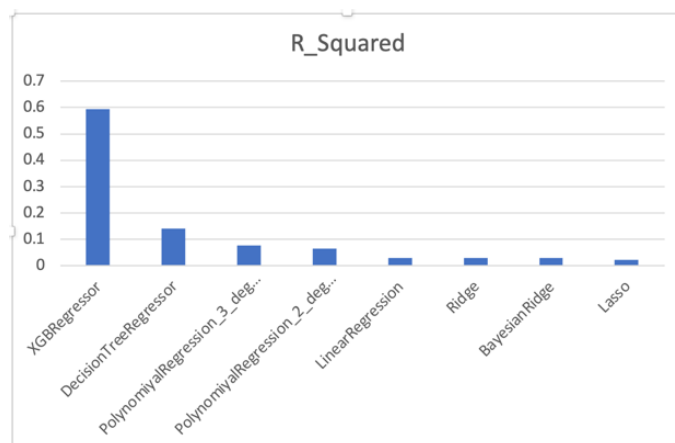


Fig. 5 R_Squared of different models (Photo credit: Original)

The definition of R_Squared tells that the more its value is closer to 1, the greater the relationship between the independent variable and the dependent variable, and the better the fit in the same situation. From Fig. 5 can learn from the table that the R_Squared of XGBRegressor is much larger than the other models and that it is the best-fitting model with a closer relationship between the dependent variable and the independent variable under this metric.

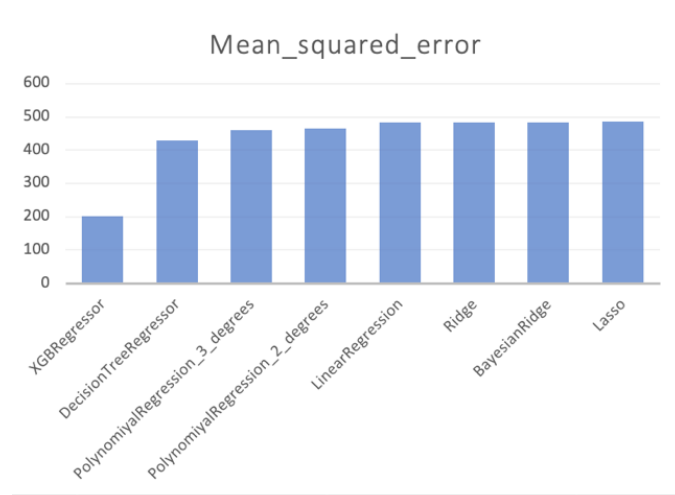


Fig. 6 Mean_squared_error of different models (Photo credit: Original)

Mean_squared_error is a variable used to calculate the error. In this study, the smaller its value, the smaller the error in the prediction results and the better the fit is. From the comparison results in Fig. 6, it is obvious that the mean_square_error value of XGBRegressor is significantly smaller than the other fitting models, and the solution can be found that XGBRegressor has the smallest fitting error and the best result

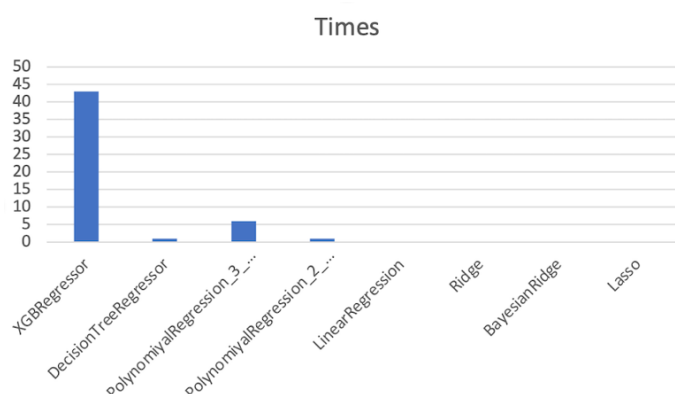


Fig. 7 Times of different models (Photo credit: Original)

Fig. 7 can see in the time comparison: XGBRegressor took 43 seconds to run, which is quite long, but based on the apparently superior performance, XGBRegressor cannot be judged to be an inefficient model - on the contrary, it is still the best

It can be found that XGBRegressor, in terms of fit, fits significantly better than the other models, both in terms of mean_squared_error and R_Squared. Although XGBRegressor took a long time to train the data, this study still consider XGBRegressor to be the best performing model based on the significantly better fit than the others.

3.3.2 Model evaluation

To evaluate the model more visually used the result of XGBRegressor training to plot the relationship between its popularity prediction results and the actual value of the popularity, which is shown in Fig. 8:

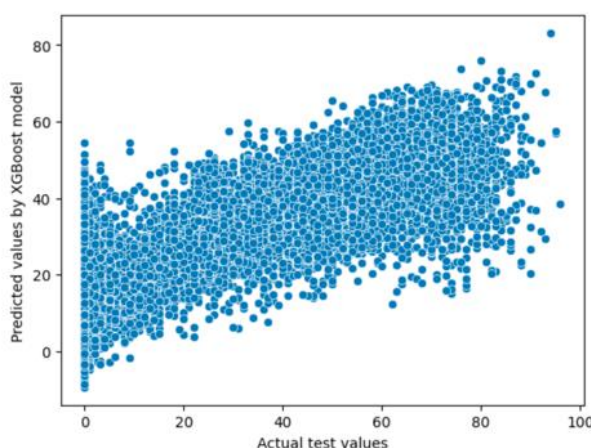


Fig. 8 Relation of predict and actual values (Photo credit: Original)

From Fig. 8 it can be found that there is a linear relationship between the training results and the prediction results, which proves that XGBRegressor is a relatively reasonable and effective model for the popularity prediction of this dataset [9].

4. Discussion

Even if the comparison finds the best model, the current prediction model still has shortcomings: the R_Squared values are still small and the average error can still be clearly detected. Given the quality of the current model, the value it provides for music popularity prediction is still quite limited. By analyzing the data processing and the data training process, this section has analyzed some of the possible reasons for the increased errors.

4.1. Values of categorical attributes have no statically significant

In the thermal coding process, this study transformed the categorical variables directly into numeric variables in the order of the data list. This transformation does provide a specific value for the attributes, but these values are unordered and not statistically significant.

In future research, it is necessary to try to give these categorical variables some values that can be evaluated to make them relevant for research. For example, for the variable artists, it is also necessary to try to calculate the mean of the popularity of all the tracks of this artist to replace the original value. In this way, the attribute artists will reflect the popularity of the artist, and famous artists tend to have more popular work, so the data will fit better.

4.2. Less attention is paid to the relationship between the independent variables.

Sometimes the combination of some independent variables may have a different effect on the popularity of music, and these should be considered in the study. It still needs to do more correlation

analysis to find the relationship between the independent variables and to explore the effect of the combination of independent variables on the dependent variable [10, 11].

5. Conclusion

The lessons that researchers are able to learn from this study are as follows.

1. By cleaning, training, and fitting the data, this study compared the effects of several training models and finally found that among the models tested, XGBRegressor was the most suitable model for predicting the music popularity of this dataset. Although it took a very long time to run, it was a much better fit than the other models, both in terms of mean_squared_value and R_Squared values. The results also suggest that the concept and algorithm of decision trees for music classification and popularity prediction is the right direction for research and that the results of this fit will be an obvious guide when using other models in the future, or when changing the parameters of the model.

2. When fitting with XGBRegressor, researchers were able to explore the effect of each attribute of the music on its popularity, but while this was effective, there is still much room for improvement. There are many other factors in this dataset that are worth exploring, such as: does combining the independent variables into a whole provide a better fit for popularity? Does the fit work better when the categorical variable is transformed into a meaningful numeric variable? The conclusions obtained from this study point to a good head start in the direction of our future research. In the future, the further study will try to clean the data in a more efficient and prediction-friendly way and try to investigate the relationship between the independent variables and the effect of their combination on the results.

References

- [1] MAHARSHIPANDYA(2022)Spotify Track Dataset <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>
- [2] Faheem khan, Ilhan tarimer, & Buse cennet karadağ . (2022 10). Effect of Feature Selection on the Accuracy of Music Popularity Classification Using Machine Learning Algorithms. Mdpi. <https://www.mdpi.com/2079-9292/11/21/3518>
- [3] Xiaoqun liao; nanlan cao; ma li; xiaofan kang. (2019). Research on Short-Term Load Forecasting Using XGBoost Based on Similar Days. IEEE. <https://ieeexplore.ieee.org/document/8669635>.
- [4] Zhagparov; zh. buribayev; s. joldasbayev ; a. yerkosova; m. zhassuzak. (2021). Building a System for Predicting the Yield of Grain Crops Based On Machine Learning Using the XGBRegressor Algorithm. IEEE. <https://ieeexplore.ieee.org/abstract/document/9465938>
- [5] Weijun su; huating yang; zaiwen liu. (2010, September). System Identification Based on Orthogonal Polynomial Regression Analysis Method. IEEE. <https://ieeexplore.ieee.org/document/5573613>
- [6] Guohao li; jiangdong wang; xiaotong jia; zijiang yang. (2021). A New Piecewise Linear Representation Method Based on the R-Squared Statistic. IEEE. Ma Kunlong. Short term distributed load forecasting method based on big data. Changsha: Hunan University, 2014.
- [7] Yunjing an, Shutao sun, & Shujuan wang. (2017). Naive Bayes Classifiers for Music Emotion Classification Based on Lyrics. IEEE. <https://ieeexplore.ieee.org/abstract/document/7960070>Fangfang. Research on power load forecasting based on Improved BP neural network. Harbin Institute of Technology, 2011.
- [8] Yang sai; ren jinxia; li zhongxia. (2009). Learning of Neural Networks Based on Weighted Mean Squares Error Function. IEEE. <https://ieeexplore.ieee.org/document/5369244>
- [9] Xiaoqun liao; nanlan cao; ma li; xiaofan kang. (2019). Research on Short-Term Load Forecasting Using XGBoost Based on Similar Days. IEEE. <https://ieeexplore.ieee.org/document/8669635>.
- [10] Junghyuk lee; jong-seok lee. (n.d.). Music Popularity: Metrics, Characteristics, and Audio-Based Prediction. IEEE. <https://ieeexplore.ieee.org/document/8327835>
- [11] Joshua s. gulmatico; julie ann b. susa; mon arjay f. malbog; aimee acoba; marte d. nipas; jennalyn n. mindoro. (2022). SpotiPred: A Machine Learning Approach Prediction of Spotify Music Popularity by Audio Features. IEEE. <https://ieeexplore.ieee.org/document/9776765>